

1. An equation is shown.

$$\frac{15}{7} + h = \frac{37}{7}$$

Complete the statement.

The value of h can be determined to be

- ☐ $\frac{15}{7}$
- ☐ $\frac{22}{7}$
- ☐ $\frac{37}{7}$
- ☐ $\frac{52}{7}$

because

- ☐ $\frac{15}{7}$
- ☐ $\frac{22}{7}$
- ☐ $\frac{37}{7}$
- ☐ $\frac{52}{7}$

is

- ☐ $\frac{15}{7}$
- ☐ $\frac{22}{7}$
- ☐ $\frac{37}{7}$
- ☐ $\frac{52}{7}$

- ☐ less than
- ☐ more than

- ☐ $\frac{15}{7}$.
- ☐ $\frac{22}{7}$.
- ☐ $\frac{37}{7}$.
- ☐ $\frac{52}{7}$.

Solve each equation.

2. $-6x = 108$

3. $147 = b - (-12)$

4. $w + 28 = 17$

5. $\frac{h}{25} = -9$

6. Buddy is planning a trip from Tampa to Jacksonville. He finds that traveling by car will take 204 minutes. He finds that traveling by train will take 112 minutes more than traveling by car.

- a. Write an equation that can be used to determine the time it takes to travel by train, x .

Equation	
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- b. Use your equation to find the time it takes to travel by train. Show your work.

7. As the world's most expensive spice, saffron provides a beautiful golden color and pungent, aromatic flavor to foods. It has been used for centuries in Persian, Arab, European, and Indian cuisine. A package of saffron costs \$4 per gram. A restaurant buys a saffron package that weighs 75 grams. Which equation can be used to determine the cost of the package, c .

- A. $c - 75 = 4$
 B. $c + 4 = 75$
 C. $\frac{c}{75} = 4$
 D. $4c = 75$

8. A small submarine stops every 2.1 miles it descends below sea level. A bar model that represents the submarine's distance in relation to sea level is shown.

-2.1	-2.1	-2.1	-2.1	-2.1
h				

Complete the statements.

The value of h in the bar model is

- ☐ -2.1.
- ☐ 5.
- ☐ -10.5.

The bar model represents the equation

- ☐ $-2.1h = -5.$
- ☐ $\frac{h}{5} = -2.1.$
- ☐ $h = -2.1.$

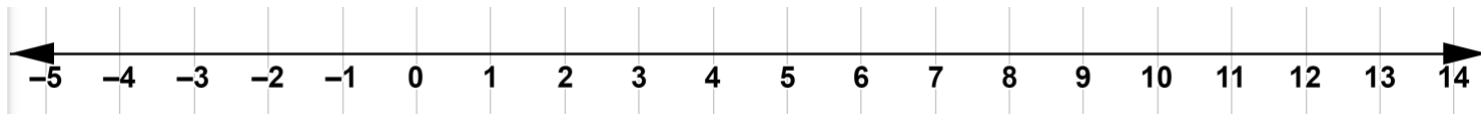
9. The Marjorie Harris Carr Cross Florida Greenway is a hiking trail that traverses Citrus, Levy, Marion, and Putnam counties. The Greenway is 110 miles long. The Florida National Scenic Trail overlaps with the Greenway for 36 miles.

Write an equation that can be used to determine the number of miles, x , the Greenway does not overlap with the Florida National Scenic Trail?

Equation	
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10. The American Red Cross recommends a minimum of 9 feet of water depth for head-first dives, including dives from pool decks.
- a. Write an inequality to represent the depth, d , in feet, that a pool should be for someone to safely dive in head-first.

- b. Represent the inequality on the number line.



Solve each inequality. Show your work. Graph the solution on the number line.

11. $-5x < 108$

12. $w + \frac{32}{3} > \frac{17}{6}$



13. $-72 \leq \frac{y}{2}$



14. For a fire hydrant to reach its maximum water flow, a building must be no farther than 76.25 meters (m) from the hydrant. Circle all of the distances that would result in a fire hydrant reaching its maximum water flow.

$$\frac{1200}{12} \text{ m}$$

$$76.3 \text{ m}$$

$$76.2 \text{ m}$$

$$74\frac{5}{9} \text{ m}$$

$$76.05 \text{ m}$$

$$76.5 \text{ m}$$

$$76\frac{1}{8} \text{ m}$$

$$\frac{7620}{1000} \text{ m}$$

$$76\frac{12}{16} \text{ m}$$

$$\frac{1500}{20} \text{ m}$$

15. Michael takes his bike to a bike shop for repair. Labor costs \$17.50 an hour. Michael's budget allows him to spend no more than \$112 on labor.
- a. Write an inequality that can be used to determine the number of possible hours, x , the bike shop can work on Michael's bike.

Inequality	
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- b. Solve the inequality.